Structural constraints and the limits of decipherment

No accepted reading is proposed. This investigation did not establish a phonetic value, identify a word, choose a language, or translate an inscription. The contribution is a reproducible set of structural tests, prediction benchmarks, photographic checks and failed-hypothesis audits.

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The strongest reproducible results

Finding	Evidence	What it establishes
Sign-order dependence	0.890 bits per noninitial sign; grouped 95% interval 0.817 to 0.965.	Transferable graphic sequencing beyond independent position and known metadata.
Second manual annotator	74 faces; +0.725 bits per candidate event.	The order result survives a different transcription, with shared catalogue/source limits.
One known missing slot	Exact top-one 33.8%, versus 17.0% baseline.	Ranked common-glyph suggestions under a verified one-slot assumption.
Unknown gap length	Count accuracy 43.0%; one-sign control 51.1%.	Context alone is unreliable for deciding how many signs were lost.
Semantic identification	No secure sound/meaning bridge established.	Useful structural advance; no substantive semantic decipherment breakthrough.

Three parallel research agents handled corpus/provenance, primary literature/novelty, and independent statistical methods. Root integrated the work and produced separate archaeological-context, paired-surface, restoration, independent-annotator and digitization experiments. Model size or confidence was never treated as evidence of a decoded language.

1. Evidence, units and source identity

Material	Scope	Status
Lipi observational export	5,680 face records; 4,695 object-family IDs. Clean RTL single-line subset 2,930; main length>=3 subset 2,107.	Derivative export associated with a disputed Sanskrit decipherment. Translations, Sanskrit, notes and text-derived class labels excluded.
Mayig photo digitization	179 Mohenjo-daro seal faces; 186 original lines; 94 explicitly clear faces.86 clear single-line faces length>=3.	Independent Michael Carlson manual annotation of CISI photographs; MIT2024. P IDs are local class IDs.
CISI1 scan	Government-hosted Joshi/Parpola 1987.862 PDF pages comprise two identical 431-page copies.	Offset 431 duplication verified for every page image. Count one book, not twice the artifacts.
Finnish 1979/1980/1982 editions	Institutional primary corpus, documentation and concordance acquired separately.	1979 and 1982 explicitly CC BY4.0.1980 license not inferred from the other editions.
Photographic audits	Initial 25-object audit; 13 selected fish/stroke cases; copper, duplicate, restoration and paired-face extensions.	Nonrandom audit samples. Compatibility counts are not population error rates.

Codes identify graphics. Wells 002 denotes two SHORT bars,032 two TALL bars; 700 a simple U,740 a handled-jar graphic. A visual count of two strokes does not establish the semantic numeral two. Mayig P IDs, Mahadevan numbers, Wells numbers and Finnish stable codes cannot be equated by their digits.

The observational code 000 denotes an eroded/illegible unit; a Mayig P000 can denote a lost section of unknown extent.999 denotes a physical blank between signs. Neither is a word delimiter or proof of one missing glyph. Complete-observed flags describe a recorded inscription, not proof that every inscription on the physical object has been inventoried.

Encoded Lipi strings are visual left-to-right lists in a presentation arranged for reading from the right. We reverse within the retained normalized single line for computational reading. Dir is preserved as a catalogue field; primary conventions use impressions for seal texts, so it does not by itself describe every physical incision's direction. Mayig likewise records impression layouts. Multiline records require separate treatment. Source definitions and exclusions are preserved in data/normalized/.

Lipi has no explicit repository license in the pinned source. Its bulk extraction remains local research material; this report and code do not license redistribution of that corpus. Pinned commits: Lipi fc8813a52e183d19c441631dc26c33a4dccbea39; Mayig ad2f1e218a34b8c33c57de0d6cb8d99272765bbb.

2. How the structural test works

The main population has 2,107 complete explicit-RTL, single-line records with at least three signs,1,596 distinct whole strings and 1,526 transitive components. A component joins all records linked by physical artifact aliases or an identical whole string. No component crosses an outer validation fold. The largest contains 36 records.

Five outer folds evaluate prediction. Three inner grouped folds select an independent baseline among global unigram, length/position and length/position/site/object-type distributions, and select smoothing for the dependent model. Selection uses training records only. Candidate smoothing values are 5,20,100,500 and 2,000. The observed selections are PM500 and PMB100 in all outer folds.

```
PMB(c | previous, length, slot, metadata)
proportional to PM(c | length, slot, metadata) * BG(c | previous) / U(c)
Normalize across the TRAINING alphabet plus UNKNOWN.
```

The test measures a reduction in held-out noninitial cross-entropy, in bits per sign. Unknown test glyphs share one descriptive UNKNOWN bucket in the likelihood calculation. Exact restoration counts such fine-code predictions as failures. Cluster-bootstrap intervals condition on the fitted cross-validation models; they do not incorporate all transcription or model-selection uncertainty.

Weighting/inventory	Predictive gain(bits/sign)	Conditional grouped 95% interval
All clean records	0.890	0.817 to 0.965
Unique whole strings, all physical aliases retained	0.848	0.807 to 0.890
Only 861->817,405->390,706->705 merged	0.930	0.862 to 0.997

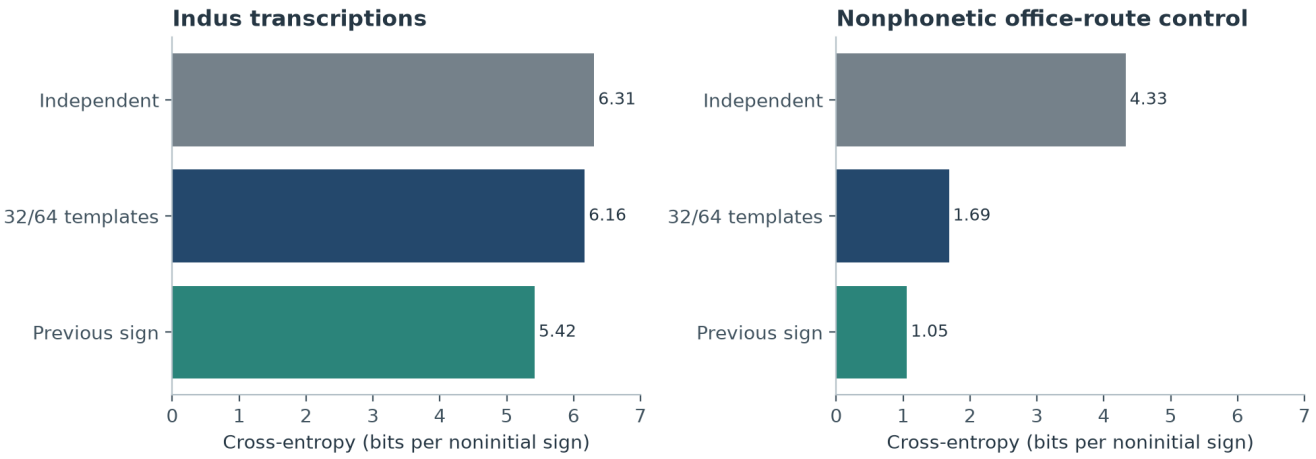
The three merges are compatibility sensitivities based on shared historical catalogue classes. They are not photographic proof of allography. Including length-two strings creates an 804-record transitive component,27.4% of the 2,930-record population; that broader design has substantially weaker effective independence.

The stronger shuffled reference rebuilds grouping/folds, selects smoothing and fits models on each full shuffled corpus while preserving exact length/position/site/type marginals. It has 1,000 completed metadata shuffles and a one-sided Monte Carlo reference p=0.000999. Its p-value is a comparison to that declared null, not a probability that the script is linguistic.

The unique-string reference additionally rejects shuffles that create duplicate whole strings. It completes 1,000 accepted draws from 6,569 proposals, rebuilding grouping/folds, tuning and prediction. Observed gain 0.848 bits/sign has zero exceedances, reference p=0.000999; null mean 0.007 bits. The accepted reference preserves the unique sampling rule and all physical aliases.

Earlier fixed-fold nulls and foldwise fixed-discovery tests remain archived and labelled as conditional diagnostics. Their apparent precision must not replace the full-procedure references. A loader defect that discarded physical aliases after uniqueness filtering was repaired before the headline unique-string and restoration results; unsafe outputs remain explicitly superseded.

3. Order is not a diagnosis of language



Lower is better within each corpus. Absolute values across corpora are not a language test. Both systems benefit from local dependencies.

Both datasets are fit using the same normalized prediction protocol. Lower cross-entropy is better within a dataset; comparing absolute values between them does not classify language.

A finite mixture of independent-slot templates, allowing up to 64 components selected on training data, improves the independent Indus baseline by 0.142 bits. Previous-sign information still improves that mixture by 0.748 bits, interval 0.688 to 0.809.

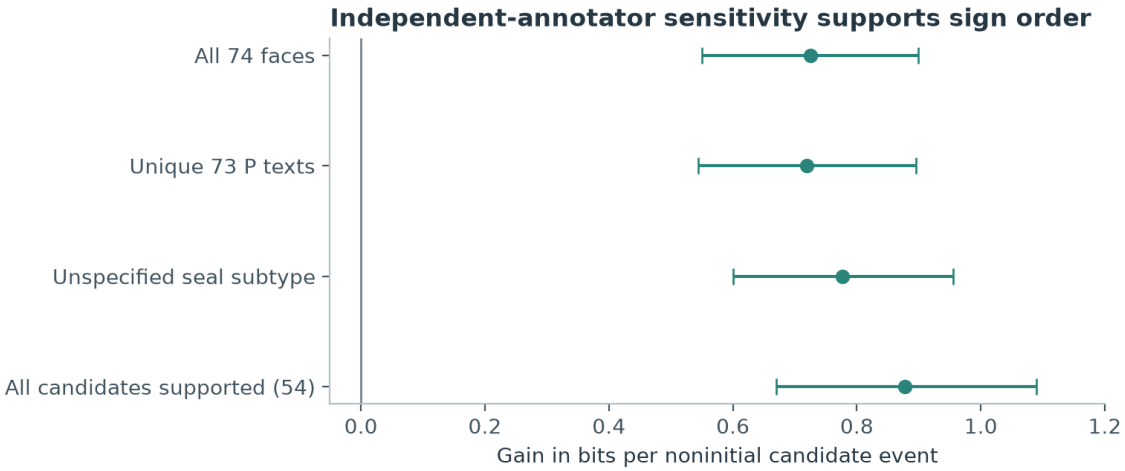
A constructive nonphonetic control contains 3,000 office-routing records. Its symbols identify office entities and legal handoffs; they encode no speech. Under the same protocol its Markov model improves the selected template family by 0.631 bits, interval 0.588 to 0.677. Thus residual adjacency dependence is not specific to phonetic writing.

This control is a counterexample to a diagnostic inference, not a historical model of Indus administration. Published critiques already make the broader point. We also acquired author-preserved real nonlinguistic data: the totem encoding has 43.7% held-out OOV tokens and very sparse observations, so its small gains are not a comparable threshold. The weather archive forms one station/text component under conservative alias closure; independent grouped cross-validation is infeasible.

Administrative templates and Sproat comparator data also appear in Nair 2026, arXiv 2604.17828 v 1. This unreviewed prior limits any claim to have introduced such controls. Our contribution is the particular normalized predictive assay and artifact/text grouping protocol.

The label-symmetry demonstration renames every fine-code state and model parameter consistently. Likelihood differences remain below 2 e-15 bits and restoration-mass differences below 6 e-18. Such an objective cannot choose among arbitrary unanchored sound/meaning labels. An independently fixed linguistic model, physical anchor or risky semantic prediction can break that symmetry; this does not make historical decipherment impossible.

4. A second manual transcription



Shared CISI sources and catalogue compatibility; overlapping candidate events are not a normalized Mayig-label predictor. Intervals condition on fitted models.

Intervals condition on fitted models. Each event is a set of possible fine Wells codes; it does not identify a single token-level graphic variant.

This test uses all 86 eligible Mayig clear single-line faces of length>=3, without selecting by agreement with Lipi. Five contain unmapped P classes; seven more have a slot with no training-supported fine candidate.74 faces are scored, with 73 distinct P strings.20 scored faces include some unsupported alternative fine candidates; the fully supported 54-face sensitivity is reported separately.

Training excludes every component touched by any CISI M1..M199 source family in the full observational register, including damaged/short/multiline records. It also excludes any complete fine-code string compatible at every position with an eligible Mayig candidate-set text. This removes 170 of 1,596 training strings, leaving 1,426 strings in 1,362 components.195 source objects are identified; 155 occur in eligible training material.

The forward recursion sums over compatible fine-code paths and normalizes after each observed candidate event. It has been checked against exhaustive path enumeration and singleton Model.score. Unseen fine candidates are not credited through UNKNOWN. Candidate sets can overlap across P labels, so this is an event likelihood, not a normalized prediction distribution over Mayig labels.

The primary stratum assumes Mohenjo-daro/SEAL:S. Replacing its seal subtype with an unseen unspecified subtype gives a similar positive gain. The annotation and crosswalk are independently supplied by Michael Carlson, but the photographs and historical catalogues are shared. This is useful replication across annotation, not an independent archaeological corpus or a new sign inventory.

A post-photo-audit sensitivity excludes the three scored faces with independently flagged segmentation/height/decorated-variant discrepancies (M81, M140, M181; a fourth flag, M62, was not scored). The remaining 71 faces give +0.755 bits per event, conditional interval 0.571 to 0.932. This check uses already observed outcomes and unchanged fitted models; it is exploratory source-quality sensitivity, not a new preregistered confirmation.

5. Restoration with a known one-sign slot

Metric	Position/site/type baseline	Local context
Exact top-one	17.0%	33.8%
True code among top five	35.5%	55.3%
True code among top ten	45.6%	62.7%
Each whole text equally weighted	17.4%	32.6%

The benchmark masks every glyph in 1,596 unique texts, totaling 8,009 slots, in 1,526 artifact/text components. Exact gain is 16.82 percentage points, conditional cluster interval 15.64 to 17.98. All outer folds select alpha 20 using grouped inner training accuracy. A separate reviewer checked a wider U/P/PM baseline family and still selected PM20.

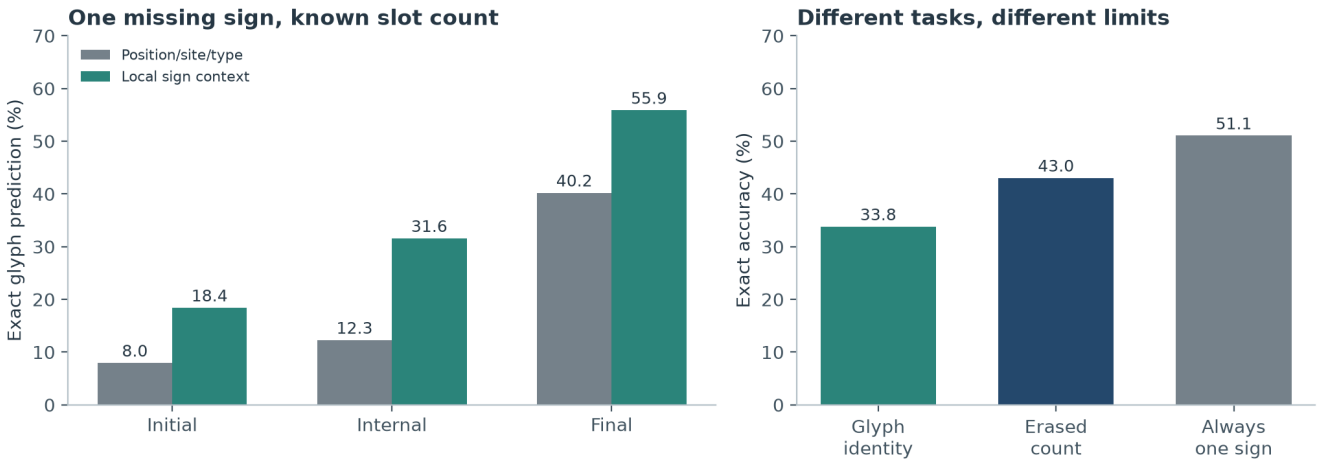
The compatible-merge sensitivity yields 35.9% exact local accuracy versus 17.8% baseline on 7,917 slots in 1,575 unique texts. This is graphic prediction; no sound or meaning is assigned. Across all 548 observed fine-code types, macro-average exact accuracy is only 5.13% local versus 0.63% baseline. The 205 singleton types cannot be recovered when unseen in training.

Entire held-out site	Unique test texts	Exact local gain(points)	Conditional 95% interval
Mohenjo-daro	866	14.94	13.53 to 16.35
Harappa	496	15.48	13.40 to 17.59
Lothal	51	17.93	10.22 to 25.40

Site exclusion is defined using every raw clean record at the held-out site, rather than only the retained unique representative. Entire touched training components are removed. Representative metadata may still discard copies at other sites, so this is a conservative transfer sensitivity, not proof of a universally accurate decoder.

The nonrandom photographic compatibility check holds out 16 fine-coded derivative texts and every linked artifact/string. Local exact accuracy is 33/79 slots, baseline 15/79. Those cases were selected for source audit, and catalogue-family compatibility is weaker than fine-glyph identity; 41.8% is not an independent population accuracy estimate.

6. Damage extent and calibrated uncertainty



Glyph:8,009 held-out slots. Count:1,596 texts with artificial internal erasures. The one-sign control applies only to the count task.

Initial/internal/final position here means reading order. The artificial gap-count task has a different truth distribution from the single-slot graphic task.

The unknown-length gap experiment removes one internal span of 1..3 signs from each held-out text, giving the predictor only survivors and site/type. Exact marginalization sums every hidden-code path. Local count accuracy is 43.0%, versus 41.9% factored baseline; paired gain 1.13 points, interval-0.19 to 2.47. Always guessing one sign scores 51.1% under this artificial erasure law.

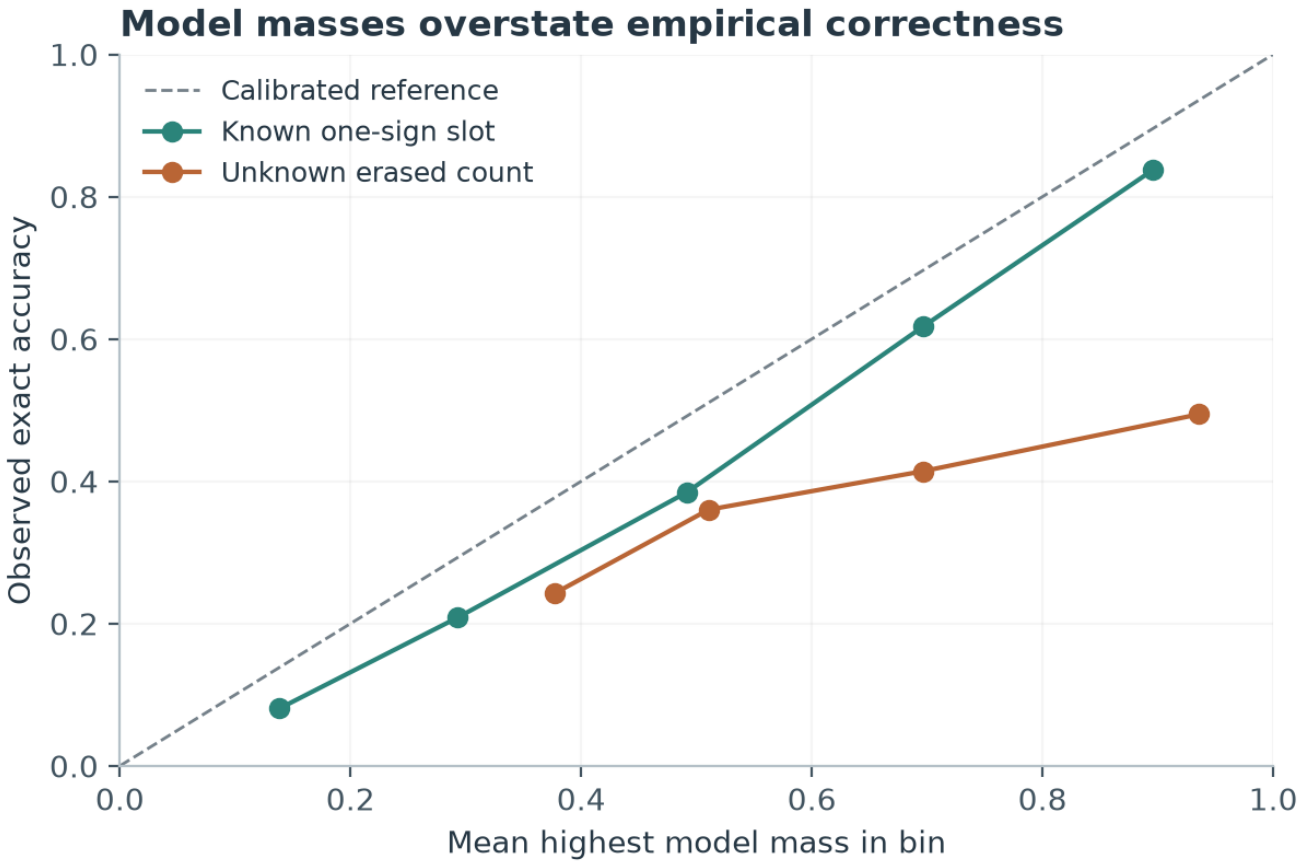
The gap model is therefore unsuccessful as a dependable count estimator. Its strong-looking masses are especially misleading: the highest bin averages 0.936 model mass but 49.4% exact correctness. The one-slot predictor is also overconfident. No model mass is presented as an archaeological confidence probability.

A reviewer identified a mismatch between the historical length prior and the artificial sampling law: the latter samples the erased count uniformly from 1 through min(3, L-2). A separate post-audit sensitivity reweights each saved count posterior by 1/min(3, L-2), leaving models, folds and survivors fixed. Independent accuracy becomes 45.61%, local accuracy 46.30%, and local gain 0.689 percentage points (conditional grouped interval -0.569 to 1.940). Local accuracy remains 4.762 points below the constant-one control (interval -7.775 to -1.726). This strengthens the negative conclusion and is not an archaeological gap prior.

A blind review of 17 Mayig faces containing one P000 found no secure new single-sign restoration. M169 initially looked promising, but its interval is wider than neighbouring signs and contains separate remnants; the historical fine-code export contains two 000 units there. That catalogue count is not ground truth, but it corroborates rejection of the one-slot premise.

A second frame of seven Lipi internal-single 000 cases yielded five acquired primary plates. M61 has a bounded partial graphic already annotated uncertain W081 by Mayig, so suggesting it would be postdiction. M62 has a visible U-like graphic already annotated W700, and a tall/short coding disagreement. These are recoding/audit cases, not newly recovered historical text.

7. Model mass and actual correctness



Each plotted point is a held-out bin mean; the diagonal is the calibration reference. These curves demonstrate empirical overconfidence in this corpus and benchmark design.

For a known one-sign slot, the 0.4..0.6 mass bin averages 0.492 model mass and 0.384 exact accuracy. The 0.8..1.0 bin averages 0.896 and 0.838. A high rank is useful for narrowing alternatives, but it must be combined with surviving carving, correct sign height, orientation, line boundaries and independently assessed slot extent.

The predictor reports alternatives rather than modifying inscriptions. It rejects 000 and 999 as proof of an exact missing slot, accepts explicit artifact exclusions, and removes every connected training component touching those artifact IDs. Source cases with an unknown number of lost signs must not be passed to the one-slot interface.

```
python src/restoration.py predict --sequence "817,002,?,740"
```

The example is an exploratory graphic query. For a source case, provide --site, --type, --alpha 20 and --exclude-artifact with the actual observational object-family ID. All-local training can contain matching texts on different objects; the output explicitly warns that exploratory suggestions may overlap known material. Accepted epigraphic restoration remains a separate human/source judgment.

8. Archaeological context: what survives

Context prediction uses observation fields only. A principal inscription per physical object is selected deterministically by greatest retained length, then record ID. Models compare sign presence/boundaries/bigrams against a site-by-length baseline. All preprocessing is fit inside grouped training folds. Text-derived structural class labels and decipherer translations are excluded.

Question	Robust result	Interpretation
Seal versus tablet, one representative per whole text	1,679 representatives; sign features improve log loss by 0.0482 nats/object; 95% interval 0.0271..0.0688; 1,999 site/length permutations p=.0005.	Some textual distribution tracks artifact function beyond these baseline covariates. It does not identify a commodity or title.
Bull 1 versus other field motifs	1,020 representatives; gain -0.0568 nats; interval -0.0792..-0.0346; p=.556.	This model fails to improve motif prediction. Absence of predictive gain is not proof of no semantic relation.
Mohenjo-daro excavation blocks	Known-period subset 316 objects; block-similarity p=.056. Unique and compatible-merge sensitivities p=.0634.. .0864.	Weak/nonrobust local association. Blocks can be trenches/streets; they are not asserted households.

Repeated whole texts can have different archaeological labels. Rowwise label permutations on the original duplicate-weighted population are difficult to interpret because duplicated-text cohorts are not independently exchangeable. The headline context sensitivity retains a deterministic source-ID representative of each whole text, records disagreements and does not choose a majority label after observing outcomes.

Artifact-type site-transfer results are mixed and sometimes lose predictive performance. Thus the in-corpus context gain cannot be promoted into a universal artifact classifier. Missing excavation periods are excluded in the stronger microcontext sensitivity; a dash is not a known shared archaeological date.

9. Paired faces and stroke modules

The query distinguishes adjacent two/three/four TALL strokes plus simple U700, in either order, from short bars and handled jar 740. It finds 316 objects with a qualifying face and a complete other inscription:83 two-stroke,123 three-stroke and 110 four-stroke.371 objects qualify overall; 55 lack a retained complete other face.

A tempting observation - no object has two unequal qualifying module counts - has almost no exposure to test that proposition.370 objects have one qualifying face; only one has two, both three-stroke. The source H355 prism has distinct B/C faces; it is not two photographs of one face. The absence of unequal counts is therefore not strong semantic exclusion evidence.

Other-face string	Objects	Opposite tall-stroke counts	Limit
176,740,400	29	Two:2; three:18; four:9	A repeated label occurs opposite varying modules; the broader family relation is published prior.
003,861	22	Three:1; four:21 in derivative codes	H2238 middle-face short bars are verified; its reverse tally is unresolved in the individual drawing.
Within-label geometry	60 objects in four site/ type/text strata	Width coefficient p=.299; Holm-adjusted .598. Height p=.603.	Dimensions are not mass or contents. Sparse negative results cannot rule out numerical readings.

Primary HARP cohort photographs support a three-versus-four long-bar contrast between manufacture groups, but lack individual excavation IDs needed to assign the photographed representative to H2238. Cohort-level corroboration must not be converted into an individually verified exception. Mould/manufacture dependence remains incompletely documented.

The paired relation challenges a deterministic mapping from the other-face label to one stroke count. An unchanged name or category could accompany varying quantities. Ration, fee, offering, entitlement, quantity and nonnumeric categories remain alternatives. No referent has been selected.

10. Substitutions, graphic classes and priority

Many attractive pairs share a coarse external catalogue class:817/861,390/405 and 705/706. These can reflect fine inventory splitting. The strongest raw pair mapped to disjoint Mayig P classes,817/820, shares eight exact whole frames; its opener environment is already published. Disjoint catalogue coding is not independent token-level photographic certification. We count distinct frames, discover on training frames, and validate in a wholly new held-out frame.

Full-procedure transfer	Observed disjoint P-class pair-frames	Metadata-null mean	1,000-shuffle p
All complete records	5	5.005	.541
Unique strings, physical aliases preserved	3	.096	.001
Three compatible-class merges	5	5.658	.639

A uniqueness-conditioned sensitivity rejects shuffled corpora that create duplicate whole strings: three transfers versus mean 0.116, p=.003 in 1,000 accepted draws from 6,569 attempts. This supports the unique-weighted aggregate constraint, while the primary and merged references remain unremarkable. It does not establish a new semantic substitution family.

The three unique disjoint P-class confirmations are 176/904,220/231 and 233/235, each in one new frame. An aggregate p-value is not an individual p-value. A post-pilot frequency-adjusted maximum test is globally positive, largely driven by known opener/fish correspondences; 176/904 has primary familywise p=.130 at 10,000 shuffles, and 220/231 p=.863.

One exact 176/904 contrast, Ad 3/H2139 in frame 235,705,033,*,740, is already diagrammed in RMRL bulletin 6(June 2025), printedp 17/PDF23 Fig 6. No earlier explicit statistical pair-class test was found in that passage, but the juxtaposed examples are prior. Fine photographic verification of all six distinct strings remains incomplete.

The initial 25-object photographic audit found catalogue compatibility in 17 of 24 eligible complete single-line comparisons. Seven differences include compound-versus-two-grid segmentation, height-category disagreement and decorated variants. These observations prevent treating a broad crosswalk as fine-glyph ground truth. The sample is not a random error audit.

11. Semantic and external-anchor audits

Fish, strokes and copper devices

Short versus tall bars are visibly distinct in the 13 selected source cases. Source checks identify opposite coding errors: Mayig assigns tall bars in M140 where the photo supports short ones; Lipi assigns tall bars in M62 where the photo supports short ones. Their glyph-height classes must be retained before testing any numerical or rebus proposal.

Published fish/star interpretations concern six/seven short-bar configurations and specific modified-fish constructions. A two-long-bar+fish Venus proposal uses a space/white rebus rather than cardinal two. A test that pools all bars as numerals would attack a different hypothesis.

Copper reverse-device families are already described in Parpola 2005/2008. Literal animal-name readings depend on fine ligature identity and on the device/label referent relation. Broad P124 merges include disputed outer leaf/person and inner feet distinctions; partial/corroded source photographs do not securely resolve C5/C6 fine identity. The observed relation does not uniquely select animal names, deities, institutions, clans or offices.

The familiar Gulf objects do not supply aligned readings

Shu-ilishu names a Meluhhan interpreter in cuneiform and depicts a scene, but the inspected object supplies no Indus sign sequence to align with that name. Gadd no 1(BM120573/U7683) has bison imagery and cuneiform alone. No 6 is a different cylinder; a later source appears to cite it in place of no 1. Gulf-style or pseudo-Indus signs without a paired known text do not establish phonetic values.

The inspected references therefore provide historical contact and language-context evidence, but no secure sign-to-sound bridge. This audit does not claim that every unexamined collection lacks an anchor, nor that a bilingual is logically required for every possible decipherment. New claims need inventory numbers, accessible originals and an explicit alignment.

12. Numerical scale and independent tests

Fuls 2020 derives a proposed 4.04 L unit from Wells's 40.4 L per-long-stroke estimate by assuming one long stroke equals ten. V=40.4*n and V=4.04*(10*n) are identical predictions for every long-stroke vessel. Those capacities alone cannot select the value ten, a decimal base or a spoken numeral. The value ten requires independently grounded mixed-sign/count evidence.

The primary original three pots are fitting cases, not external validation. H370 shows seven long bars without U; H371 six plus U; H372 three plus U. Published volume estimates depend on geometric reconstruction and direct dimensions rather than measured ancient contents. Funerary or secondary use does not establish what the earlier marks quantified. Later pots differ in marking inventory, units and reconstruction uncertainty; none is silently forced into the same frozen unit model.

Source fit case	Long strokes	Indexed profile volume	Eligibility limit
H372	3	About 121 litres	Profile-to-object table alignment remains provisional.
H371	6	About 241 litres	Part of original scale fitting.
H370	7	About 284 litres	No U; not an unqualified test of a U-conditioned proposal.

The indexed estimates closely reproduce 40.4 litres per long stroke, but original Table 1 PDF layout was inaccessible and excavation-number links disagree: CISI assigns 12908 to H370, whereas the indexed Fuls table associates it with H372. H95-2661 has four rim strokes in the inspected cluster, yet its excavation report describes four inscriptions and later reuse as a sump. K90 has uncertain profile units. No independently eligible held-out vessel test was established. Complete source fields, geometric alternatives and unresolved mappings are retained in research/literature/metrology/.

13. Numerical constraints: exact conditional result

A proposed reading is stronger when its assumptions imply a finite set of checkable values. We independently solved the two equalities used in Fuls's numerical argument, without a bounded search or an assumed decimal base. Let L and S be the multipliers assigned to selected long/short stroke positions and v the proposed crescent value. These are hypothetical quantities, not unconditional decoded values for every stroke.

```
v = 2*S - L
v*L + 1 = 7*S + 5
Eliminate v: (2*L - 7)*S = L*L + 4
Let d = 2*L - 7: d must divide 65.
```

All positive integer solutions	L	S	v
Solution 1	4	20	36
Solution 2	6	8	10
Solution 3	10	8	6
Solution 4	36	20	4

Conditional on either cited range 8<=L<=13 or the broader8<=L<=18, only(L,S,v)=(10,8,6) survives. The range and equalities carry the identifying information. Excluding primes or assuming S even is unnecessary; the first equality by itself only constrains the parity of v+L. Without the upper bound, L>=10 still permits(36,20,4). The exact proof is executable in src/methods/audit_numerical_equalities.py; hypothetical two/three/four-stroke inputs are compared separately in numerical_count_delta.py.

The bangle range is not independently authenticated for the object captioned M-1383 in the author's Fig 15. The original Vidale 1987 chapter (printed 105-111/PDF108-114) describes a fused DK-B, C block whose identity with that object has not been established. Its ten-bangle reconstruction is conjectural and supplies no 82-fragment/15-20 total here. The current M-1383 export has clay-tag metadata but compatible seven-short/one-long signs; the author figure/catalogue must resolve this context conflict. The range is unverified, rather than disproved.

An original object check, with the identifier gap preserved

The author's historical ICIT3786 number points to a different object in the pinned current export. Searching complete opposing sign patterns instead identifies a unique candidate, current family 1786/CISI H904. Vats 1940 Plate XCVI no 477 (PDF 200 of the plates scan) shows three long strokes between U and crescent on one face, and two long strokes between jar and overlapping circles on the other. The original table, printed page 349/PDF 378 of the text scan, gives find J213, mound D, and 0.65 by 0.28 inches, matching H904's J213(D)477 and 16.5 by 7.1 mm.

This establishes the H904-to-Vats object join and supports the candidate's coarse three/two stroke counts. The historical 3786-to-H904 join remains provisional because the author's Figure 16 image was not independently retrieved. The prose's W032/three-stroke discrepancy may be a code typo, rather than evidence for changing the coefficient to two. Extra signs and equality/addition/unit assumptions remain unresolved; source-count agreement alone does not validate the numerical semantics.

Source pagination: the 2020 proceedings chapter is pp 57-90; the inspected 2023 author reprint is pp 15-48. The first equality is on reprintp 30/PDF16, and the M331 equality on reprintp 32/PDF18. The M331 photograph supports coarse stroke configurations, but positional multiplication and equal total quantity across faces are interpretive premises.

14. A primary chart and its transcription limits

A local searchable atlas now provides all 687 chart cells from the independently acquired 1979 corpus, with original context crops, ranks, stable Finnish numeric codes and explicit alternatives. 16 cells retain ambiguous numeric readings. Source underlines and damage markers are modern editorial conventions. AI visual inspection is not expert certification, and no chart-derived inscription was added to the analytical corpus.

The initial blind numeric audit found only 14/28 clear labels correct by OCR and 15/28 by a source-font recognizer. Manual reinspection subsequently corrected digit readings; those later corrections do not revise the earlier score. Duplicate-code alerts were used to find source crops, never to force uniqueness. Historical inputs and candidate unions remain archived.

A ten-row image development pilot tested four predeclared segmentation penalties. A source-observed crop defect included neighbouring number ink and was repaired; one row was partially disclosed during that repair. At penalties 0.5 and 1, all ten drawn-unit counts matched, but first-choice candidates matched only two whole rows. This was development evidence, with the partial disclosure recorded.

Fresh twenty-row validation: reference withheld until freeze

Frozen endpoint	Result	Meaning
Modern source drawn units	96 total: 91 charted, 5 editorial	No inference of ancient text length.
Predicted units / matching counts	107 / 14 of 20 rows	Counts do not certify boundaries.
First image-template candidate	69 summed edits; 2 whole rows compatible	Numeric alternatives can widen one candidate.
First five image templates	60 summed edits; 5 whole rows compatible	Mean candidate-code set width 5.04, maximum 8.

Twenty randomly selected printed rows used the fixed penalty 1 chosen from development. All selected rows were retained, including two auxiliary-only rectangles and four rows with loss or inaccessibility marks. The manual reference was withheld until a complete prediction file existed. These scores use the chart snapshot frozen at prediction time, with 14 numeric ambiguities; the later 16-ambiguity atlas is not retroactive validation. Strict scores use set intersection in Levenshtein alignment; zero edits mean candidate compatibility, not a uniquely certified fine-code transcription.

A subsequent source-only crop audit found 18 of 20 crops entered the target row's upper ink, and 14 included following-row ink. The scores therefore assess the failed frozen acquisition pipeline, rather than uncontaminated shape recognition alone. The deliverable is a verifiable chart reference and documented acquisition limits, not a completed independent sign corpus. Original physical side/line boundaries remain unknown.

A separate post-reference geometry diagnostic retains the identical frozen chart, matcher, penalty 1 and horizontal bounds. Clean source-only vertical crops yield 96 predicted units and 18/20 matching row counts. Strict first-template candidates give 24 summed edits and 6 compatible whole rows; five templates give 6 edits and 16 rows (maximum 9 code candidates per slot). This isolates a substantial geometry defect, but is development after reference release and needs fresh validation.

15. What advances and what does not

Contribution	Status
Artifact/text alias preservation, stronger full-procedure nulls, second-annotator event replication	A reproducible methodological robustness contribution. Not first discovery of Indus order.
Known-slot restoration with train-only tuning, site exclusion, rarity and calibration audits	A useful benchmark/tool. Markov restoration itself is published prior(Rao 2009).
Unknown gap-length estimator	Failed exploratory extension. It must not be promoted to accepted restoration.
Photographic coding/extent discrepancies, scan duplication and corrigenda	Concrete source corrections/flags; sampled scope is explicit.
Opening/terminal/fish substitution categories and paired count modules	Largely replicate published findings; no priority claim.
Language identification, phonetic assignments, lexical meanings or authenticated translation	Not achieved. No substantive semantic decipherment breakthrough is claimed.
Conditional numerical proof and original H904/Vats source join	All integer solutions derived; key stroke counts corroborated. Semantics and historical index join remain conditional.
Independent 1979 digitized sign corpus	Not completed. Manual chart delivered; frozen acquisition failed. Geometry repair improves development scores and needs fresh validation.

Shared-context and substitution discovery itself is established prior, including Khanna and Merriam 2025. The contribution here is the specific alias-preserving evaluation, held-out annotation test, stronger null protocol and source-level failure accounting. No published proposed functional or semantic reading is adopted merely because it uses similar statistical methods.

The project preserves unsuccessful tests and superseded defects rather than combining their best numbers. The evidentiary hierarchy is physical carving/source identity, observational transcription, tested distributional constraint, then a separately falsifiable semantic or phonetic hypothesis. Passing the first three stages does not automatically pass the fourth.

What would make the next claim substantive

A decoder should commit in advance to exact graphic values, segmentation and an external referent/phonological model, then predict an independently acquired inscription or archaeological measurement that competing readings cannot match by freely changing units or dictionaries. Full artifact/mould aliases and every inscription on a multi-marked object must be inventoried before scoring. A new reading should also explain its failed cases.

16. Reproduce and inspect

The workspace contains original acquisitions/provenance, normalized observational files, preregistrations, exploratory plans, code, held-out audits, machine-readable result JSON and agent reviews. No cloud publication or external message was sent. Bulk derivative data remain local.

Purpose	Entry point
Integrity checks	python src/verify_root.py python src/methods/verify_methods.py
Known-slot benchmark	python src/restoration.py benchmark
Site-held-out restoration	python src/restoration.py crosssite --out results/restoration_crosssite
Independent manual annotation	python src/independent_annotator.py
Unknown-length negative benchmark	python src/gap_restoration.py benchmark
Unanchored label symmetry	python src/identifiability_demo.py
Frozen image-test evaluation	python src/evaluate_font_fresh20.py
Conditional numerical constraints	python src/methods/audit_numerical_equalities.py
Scientific figures / report	python src/research_figures.py python src/research_report.py --final
Long structural/null procedures	research/methods/REPRODUCING.md; source seeds, hashes and completion states in RESULT_INDEX.json.

Commands run from the workspace root. Python numerical, statistical, image, PDF and plotting dependencies are recorded in the final environment manifest. Some long null procedures take around an hour on this host; exact runtime is saved, and resumed seeds are preserved. The final flag refuses to generate a final edition before four elapsed hours or before the stronger 1,000 metadata reference is complete.

Important review files: research/methods/FINDINGS.md; root_restoration_audit.json; independent_annotator_audit.json; research/literature/evidence_and_novelty_matrix.md; external_phonetic_anchor_audit.md; restoration_review.md; research/corpus/primary/1979_extraction_assessment.md; plate_audit/restoration_slots/; and the final output/evidence_manifest.json.

The expanded literature matrix records exact source locations, retrieval failures,2026 preprint status and corrected source identities. Speculative projects are novelty comparators, not accepted decipherments. Acquiring a source does not endorse its reading.

Selected primary references

Koskenniemi & Parpola 1979, Corpus of texts in the Indus script
Stable-code/rank, uncertainty and printing conventions. Official CC BY4.0 entry; later documentation revisions audited separately.

Joshi & Parpola 1987, CISI1
Primary photographs; duplicate scan and corrigenda audit. Printed page numbers differ from PDFindices.

Rao et al.2009, A Markov model of the Indus script
Prior sequence prediction and damaged-text restoration. Present results add scope-specific robustness safeguards.

Sproat 2010, Last Words
Critique of structural evidence as a linguistic diagnostic. Corrected source identity; J10-4009 was unrelated.

Yadav 2013, Sensitivity of Indus Script to Site and Type of Object
Paired labels/modules and archaeological context.

Yadav 2019, statistical/structural overview
p 127 Table 1 quantifies prior opening pairs.

Mahadevan 1982, Terminal ideograms
pp 313-314 published terminal correspondence; proposed meanings remain hypotheses.

Mahadevan 1986, Towards a grammar of the Indus texts
Short markers, modified fish and grammatical proposals.

Parpola 2005, Study of the Indus script
pp 55-59 fine copper-related ligature variants.

Parpola 2008, Copper tablets from Mohenjo-daro
Device/label group relations; fine photographic limits remain.

Wells 2008, Airavati contribution
pp 285-286 published U/stroke group and mould-duplication cautions.

RMRL bulletin 6, June 2025
p 17 Fig 6 already juxtaposes Ad 3/H2139 candidate frame; commodity interpretation not adopted.

Gadd 1932, Seals of Ancient Indian style found at Ur
Primary Gulf objects and cuneiform-only/Indus-object distinctions.

Possehl 2006, Shu-ilishu's cylinder seal
Named interpreter; no aligned Indus written sequence on inspected object.

Fuls 2020 chapter,2023 author reprint, Structural Analysis of Numerical Indus Inscriptions
Inspected author posting pagination 15-48; 2020 proceedings chapter 57-90. Numerical semantics remain assumptions.

Vats 1940, Excavations at Harappa, plates
PlateXCVI no 477: original H904 candidate. Text table and dimensions independently joined; government catalogue title roles were checked against the scans.

Vidale, More Evidence on a Protohistoric Ceramic Puzzle, InterimReportsVol 2
Original pp 105-111: conjectural saggar arrangement for a fused block, not an independent M1383 total.

Khanna & Merriam 2025, A Computational Analysis of the Indus Script: Identifying Sign Functions in Logo-Syllabic Writing Systems
Prior shared-context and substitution clustering. Proposed meanings are not accepted anchors.

Fuls 2012, Volumenmaße von Tongefäßen der Induskultur
Author-posted indexed text locates measurements and modelling. Original PDFlayout unavailable; table alignment flagged provisional.

Mayig observational source, pinned 2024 annotation
Independent manual photo annotation; local P code/schema conventions, MIT.

Lipi observational export, pinned revision
Only observational fields used. Disputed translations excluded; bulk license absent.